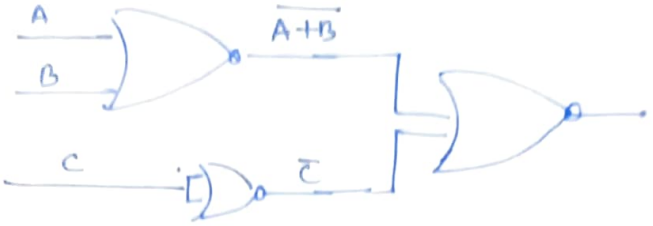
step-3



QUINE – McCLUSKEY METHOD (TABULAR METHOD)

Step-1: List all minterms in binary form.

Step-2: Separate the no. of groups according to the no. of 1's

Step-3: Compare each binary number with every group in the adjacent next higher category group and they differ only one bit position. Put check mark if comparison is possible (-) and copy the remaining term in the next column. Put (V) mark for every comparison.

Note:- The essential prime implicants are identified if they have no tick mark

Step-4: Apply the same process described in step-3 for the resultant column & continue the cycles until a single pass through cycle yields further elimination of literals.

Step-5: for Prime implicant chart

a) The prime implicants should be represented in rows & each minterm of the function in a column.

b) Crosses (x) should be placed in each row to show the composition of minterm that makes the prime implicant

(C) A completed prime implicants table should be inspected for columns containing only a single cross. Prime implicants that cover minterms with a single cross in their columns are called essential prime implicants.

Step-6: Getting the simplified expression after the above step.

Q. Simplify using tabular method.

$$f(a,b,c,d) = \sum m(0,1,2,5,6,7,8,9,10,14)$$

		a b c d
Group 0	0	0000 ✓
Group 1	1	0001 ✓
	2	0010 ✓
	8	1000 ✓
Group 2	5	0101 ✓
	6	0110 ✓
	9	1001 ✓
	10	1010 ✓
Group 3	7	0111 ✓
	14	1110 ✓

Step-1

~~Group 0~~

Group 0	0, 1	000 – ✓
	0, 2	00 – 0 ✓
	0, 8	–000 ✓
Group 1	1, 5	0 – 01 ✓ $\bar{A}\bar{C}D$
	1, 9	–001 ✓
	2, 6	0 – 10 ✓
	2, 10	–010 ✓
	8, 9	100 – ✓
	8, 10	10 – 0 ✓
Group-2	5, 7	01 – 1 ✓ $\bar{A}BD$
	6, 7	011 – ✓ $\bar{A}BC$
	6, 14	–110 ✓
	10, 14	1 – 10 ✓

Step-2

Group-0	0, 1, 8, 9	− 0 0 −	$\bar{B}\bar{C}$
	0, 2, 8, 10	− 0 − 0	$\bar{B}\bar{D}$
Group-1	0, 8, 1, 9	− 0 0 −	$\bar{B}\bar{C}$
	0, 8, 2, 10	− 0 − 0	$\bar{B}\bar{D}$
Group-3	2, 6, 10, 14	− − 1 0	$C\bar{D}$
	2, 10, 6, 14	− − 1 0	

Answer

$\bar{B}\bar{C} + C\bar{D} + \bar{A}BD$

$\bar{B}\bar{C}$, $C\bar{D}$ — essential prime implicant

$\bar{A}BD$ is also included in answer as (5,7) minterms were left & not included in essential implicant (Mark 'x' according to the prime implicant no)

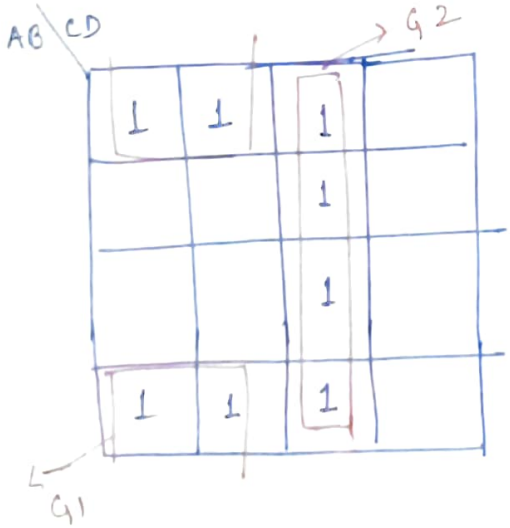
Essential Prime implicants are those who have single 'x' in a column.

Prime Implicants		0	1	2	5	6	7	8	9	10	14
1, 5	$\bar{A}\bar{C}D$		x		x						
5, 7	$\bar{A}BD$				x		x				
6, 7	$\bar{A}BC$					x	x				
0, 1, 8, 9	$\bar{B}\bar{C}$	x	x					x	x		
0, 2, 8, 10	$\bar{B}\bar{D}$	x		x				x		x	
2, 10, 6, 14	$C\bar{D}$			x		x				x	x

Prime Implicant :- → group of 1 → largest possible group of 1's

Essential Prime Implicant :-

Q $Y(A,B,C,D) = \sum m(0,1,3,7,8,9,11,15)$



$Y = G1 + G2$

$= \bar{B}C + CD$

Step 1

Group	Minterm	Binary Representation	
Group-0	m_0	0 0 0 0	No one's
Group-1	m_1	0 0 0 1	One 1
	m_8	1 0 0 0	
Group-2	m_3	0 0 1 1 ✓	Two 1's
	m_9	1 0 0 1 ✓	
Group-3	m_7	0 1 1 1 ✓	Three 1's
	m_{11}	1 0 1 1 ✓	
Group-4	m_{15}	1 1 1 1 ✓	four 1's

Step-2

Group	Minterm	Binary Representation	
Group-0	m_0, m_1	0 0 0 –	✓
	m_0, m_8	– 0 0 0	✓
Group 1	m_1, m_3	0 0 – 1	✓
	m_1, m_9	– 0 0 1	✓
	m_8, m_9	1 0 0 –	✓
Group 2	m_3, m_7	0 – 1 1	✓
	m_3, m_{11}	– 0 1 1	✓
	m_9, m_{11}	1 0 – 1	✓
Group-3	m_7, m_{15}	– 1 1 1	✓
	m_{11}, m_{15}	1 – 1 1	

Q: Realize the following logic function in SOP form using Quine McCluskey Method.

$$f(A,B,C,D) = \prod M(2,7,8,9,10,12)$$

$$f(A,B,C,D) = \prod M(2,7,8,9,10,12)$$

$$= \sum m(0,1,3,4,5,6,11,13,14,15)$$

Group	Minterms	Binary Representation	
G-0	0	0 0 0 0 ✓	Zero 1's
G-1	1	0 0 0 1 ✓	One 1's
	4	0 1 0 0 ✓	
G-2	3	0 0 1 1 ✓	
	5	0 1 0 1 ✓	Two One's
	6	0 1 1 0 ✓	
G-3	11	1 0 1 1 ✓	
	13	1 1 0 1 ✓	Three 1's
	14	1 1 1 0 ✓	
G-4	15	1 1 1 1 ✓	four 1's

Step-2

Group	Minterm Pair	Binary Rep	
0	0, 1	0 0 0 -	✓
	0, 4	0 - 0 0	✓
1	1, 3	0 0 - 1	← $\bar{A}\bar{B}D$
	1, 5	0 - 0 1	✓
	4, 5	0 1 0 -	✓
	4, 6	0 1 - 0	← $\bar{A}B\bar{D}$
2	3, 11	- 0 1 1	→ $\bar{B}CD$
	5, 13	- 1 0 1	→ $B\bar{C}D$
	6, 14	- 1 1 0	→ $BC\bar{D}$

Group-3	11, 15	1 - 1 1	→ ACD
	13, 15	1 1 - 1	→ ABD
	14, 15	1 1 1 -	→ ABC

Step-4

Group-0	0, 1, 4, 5	0 - 0 -	$\bar{A}\bar{C}$
	0, 4, 1, 5	0 - 0 -	

Preparing Table

		0	1	3	4	5	6	11	13	14	15
$\bar{A}B\bar{D}$	1, 3		X	X							
$\bar{A}B\bar{D}$	4, 6				X		X				
$\bar{B}CD$	3, 11			X				X			
$B\bar{C}D$	5, 13					X			X		
$BC\bar{D}$	6, 14						X			X	
ACD	11, 15							X			X
ABD	13, 15								X		X
ABC	14, 15									X	X
$\bar{A}\bar{C}$	0, 1, 4, 5	X	X		X	X					

$f(A,B,CD)\ \bar{A}\bar{C} + \bar{B}CD + BC\bar{D} + ABD.$

After essential prime implicant, the remaining minterms are 3, 6, 11, 13, 14, 15, so we need to find the terms which cover these minterms.